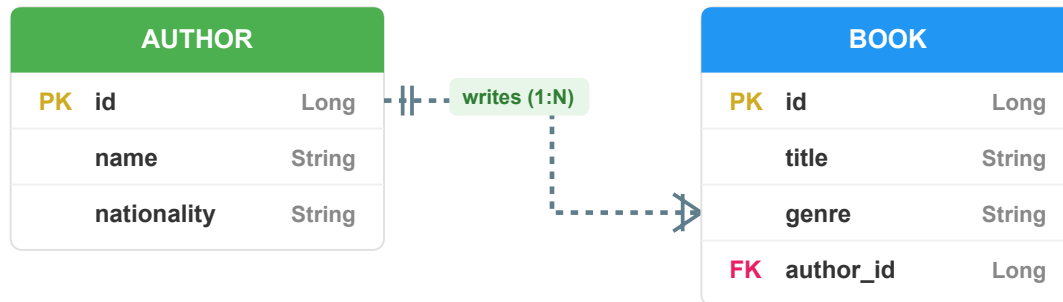


Library Management System - Database Assignment

1. Entity Relationship Design

An Author can have multiple Books (One-to-Many), and each Book belongs to one Author (Many-to-One). This relationship is strictly enforced using JPA's `@OneToMany` and `@ManyToOne` annotations.



2. Implementation Details (Code)

a) Populate Database

```
@PostConstruct
@Transactional
public void populateDatabase() {
    if (authorRepository.count() == 0) {
        for (int i = 1; i <= 10; i++) {
            Author author = new Author("Author " + i, "Nationality " + i);
            authorRepository.save(author);
            Book book = new Book("Book Title " + i, "Genre " + i, author);
            bookRepository.save(book);
        }
    }
}
```

b) Create Operation

```
@PostMapping("/addBook")
public String addBook(@ModelAttribute("book") Book book, Model model) {
    try {
        libraryService.saveBook(book);
        return "redirect:/";
    } catch (Exception e) {
        model.addAttribute("error", "Error adding book: " + e.getMessage());
        model.addAttribute("authors", libraryService.getAllAuthors());
    }
}
```

```

        return "addBook";
    }
}

```

c) Read Operation

```

// BookRepository.java (Custom inner join)
@Query("SELECT b FROM Book b JOIN b.author a")
List<Book> findAllBooksWithAuthors();

// index.jsp (Iterating over entities)
<c:forEach var="book" items="${books}">
    <tr>
        <td>${book.id}</td>
        <td>${book.title}</td>
        <td>${book.genre}</td>
        <td>${book.author.name}</td>
    </tr>
</c:forEach>

```

d) Update Operation

```

@GetMapping("/updateBook/{id}")
public String showUpdateBookForm(@PathVariable("id") Long id, Model model) {
    Book book = libraryService.getBookById(id);
    model.addAttribute("book", book);
    model.addAttribute("authors", libraryService.getAllAuthors());
    return "updateBook";
}

```

3. Screenshots

Library Management System

Books

Add New Book

ID	Title	Genre	Author	Nationality	Actions
1	Book Title 1	Genre 1	Author 1	Nationality 1	Edit
2	Book Title 2	Genre 2	Author 2	Nationality 2	Edit
3	Book Title 3	Genre 3	Author 3	Nationality 3	Edit
4	Book Title 4	Genre 4	Author 4	Nationality 4	Edit
5	Book Title 5	Genre 5	Author 5	Nationality 5	Edit
6	Book Title 6	Genre 6	Author 6	Nationality 6	Edit
7	Book Title 7	Genre 7	Author 7	Nationality 7	Edit
8	Book Title 8	Genre 8	Author 8	Nationality 8	Edit
9	Book Title 9	Genre 9	Author 9	Nationality 9	Edit
10	Book Title 10	Genre 10	Author 10	Nationality 10	Edit

Authors

Add New Author

ID	Name	Nationality	Actions
1	Author 1	Nationality 1	Edit
2	Author 2	Nationality 2	Edit
3	Author 3	Nationality 3	Edit
4	Author 4	Nationality 4	Edit
5	Author 5	Nationality 5	Edit
6	Author 6	Nationality 6	Edit
7	Author 7	Nationality 7	Edit
8	Author 8	Nationality 8	Edit
9	Author 9	Nationality 9	Edit
10	Author 10	Nationality 10	Edit

localhost:8080/addBook

Add New Book

Title:

Genre:

Author:

-- Select Author --

Save Book

[Back to Home](#)

10	Book Title 10	Genre 10	Author 10	Nationality 10	Edit
11	Atomic Habits	Self Help	Author 3	Nationality 3	Edit

Update Book

Title:

Book Title 10

Genre:

Genre 10

Author:

Author 10

▼

Update Book

[Back to Home](#)

8	Book Title 8	Genre 8	Author 8	Nationality 8	Edit
9	Book Title 9	Genre 9	Author 9	Nationality 9	Edit
10	Attitude is Everything	Self Help	Author 2	Nationality 2	Edit
11	Atomic Habits	Self Help	Author 3	Nationality 3	Edit

4. Challenges Faced & Solutions

Challenge 1: JSP Resolution in Spring Boot 3

Issue: Spring Boot 3 migrated to Jakarta EE, making traditional javax.servlet JSTL dependencies incompatible.

Solution: Updated dependencies to jakarta.servlet.jsp.jstl-api and configured the view resolver.

Challenge 2: Exception Handling for Constraints

Issue: Invalid data submissions caused the application to crash completely.

Solution: Implemented global exception handling using @ExceptionHandler to gracefully catch exceptions.

Challenge 3: Inner Join in Spring Data JPA

Issue: Fetching Books with their Authors natively could lead to N+1 query problems.

Solution: Explicitly wrote an inner join using a custom JPQL @Query to optimize retrieval.

5. Github URL

[https://github.com/\[your-username\]/LibraryManagementSystem](https://github.com/[your-username]/LibraryManagementSystem)